

TOWERS OF HANOI - RECURSIVE ALGORITHM

Java Code

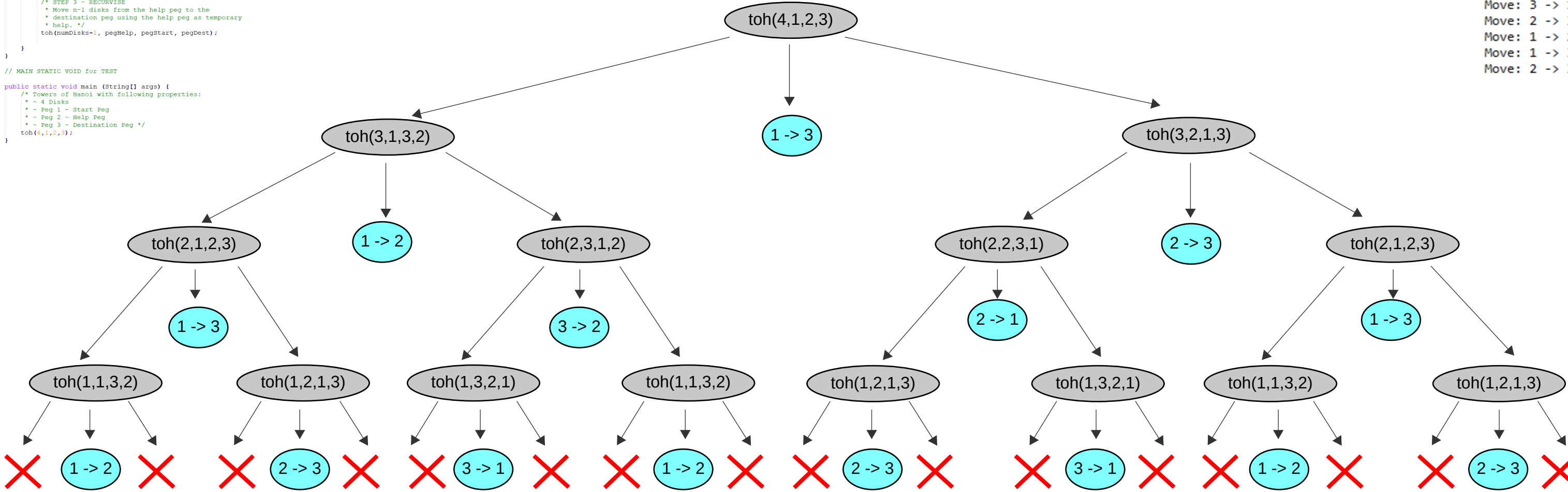
```
public class TowersOfHanoi {  
    // MAIN RECURSIVE METHOD  
    public static void toh(int numDisks, int pegStart, int pegHelp, int pegDest) {  
        // If there are still disks to be moved...  
        if (numDisks>0) {  
            /* STEP 1 - RECURSIVE  
             * Move n-1 disks from the start peg to the help peg,  
             * using the destination peg as temporary help. */  
            toh(numDisks-1, pegStart, pegDest, pegHelp);  
  
            /* STEP 2 - NON RECURSIVE  
             * Move the left disk on the start peg to the  
             * destination peg */  
            System.out.println("Move: " + pegStart + " -> " + pegDest);  
  
            /* STEP 3 - RECURSIVE  
             * Move n-1 disks from the help peg to the  
             * destination peg using the help peg as temporary  
             * help. */  
            toh(numDisks-1, pegHelp, pegStart, pegDest);  
        }  
    }  
  
    // MAIN STATIC VOID for TEST  
    public static void main (String[] args) {  
        /* Towers of Hanoi with following properties:  
         * - 4 Disks  
         * - Peg 1 - Start Peg  
         * - Peg 2 - Help Peg  
         * - Peg 3 - Destination Peg */  
        toh(4,1,2,3);  
    }  
}
```

4 Disks
Peg 1 - Start
Peg 2 - Helper
Peg 3 - Destination

Output of Java Code

Move: 1 -> 2
Move: 1 -> 3
Move: 2 -> 3
Move: 1 -> 2
Move: 3 -> 1
Move: 3 -> 2
Move: 1 -> 2
Move: 1 -> 3
Move: 2 -> 3
Move: 2 -> 1
Move: 3 -> 1
Move: 2 -> 3
Move: 1 -> 2
Move: 1 -> 3
Move: 2 -> 3

RECURSIVE CALLS TREE DIAGRAM with OUTPUTS



SEQUENCE OF MOVEMENTS REQUIRED

